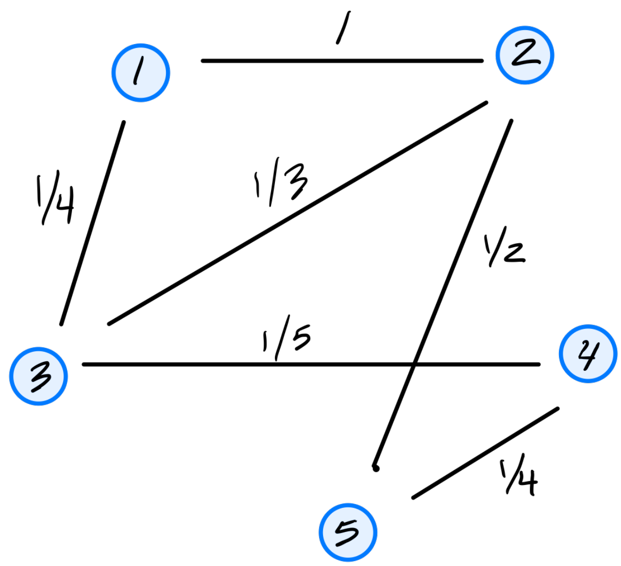
DSC 40B - Sample Midterm 02

Note: This sample midterm is intended to give you an idea of the format of the exam, but it's not intended to be a comprehensive review of the material. Also, note that this sample exam is from a previous iteration of the course, and topics can change slightly from quarter to quarter depending on the instructor and how much was covered in lecture – you should expect Midterm 02 to cover the content from *this quarter*'s Lecture 09 through 15.

You should focus your studying on the practice problems found at `https://dsc140b.com/practice` that are labeled with `midterm-02`, as those will be the most representative of the types of questions you can expect to see on the exam.

Problem 1.

Consider the similarity graph shown below.



a) What is the (1,3) entry of the graph's (unnormalized) Laplacian matrix?

b) Consider the embedding vector $\vec{f} = (1, 2, -1, 4, 8)^T$. What is the cost of this embedding with respect to the Laplacian eigenmaps cost function?

Note: as mentioned in the Campuswire post from earlier this week, there is some ambiguity in the cost function as to whether each edge should be counted twice or just once. Use the version of the cost function which counts each edge only once.

Problem 2.

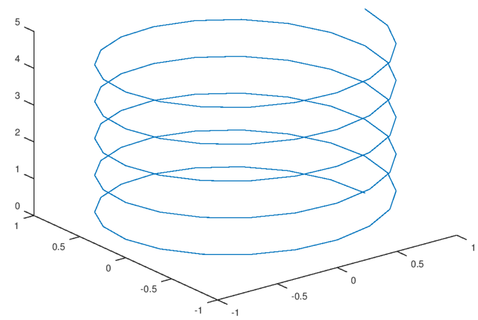
Suppose L is a graph's Laplacian matrix and let $\vec{f}$ be a (nonzero) embedding vector whose embedding cost is zero according to the Laplacian eigenmaps cost function.

True or False: $\vec{f}$ is an eigenvector of L.

- ◯ True
- ◯ False

Problem 3.

Consider the plot shown below:

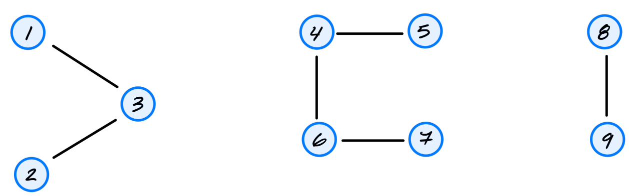


a) What is the ambient dimension?

b) What is the intrinsic dimension of the curve?

Problem 4.

Suppose a k-nearest neighbors graph is constructed and found to have 3 connected components:



Suppose $\vec{u}$ is a normalized eigenvector of the graph's Laplacian matrix with eigenvalue of zero. Suppose $u_1 = \frac{1}{\sqrt{6}}$ and $u_4 = 0$. What is $|u_8|$?

Problem 5.

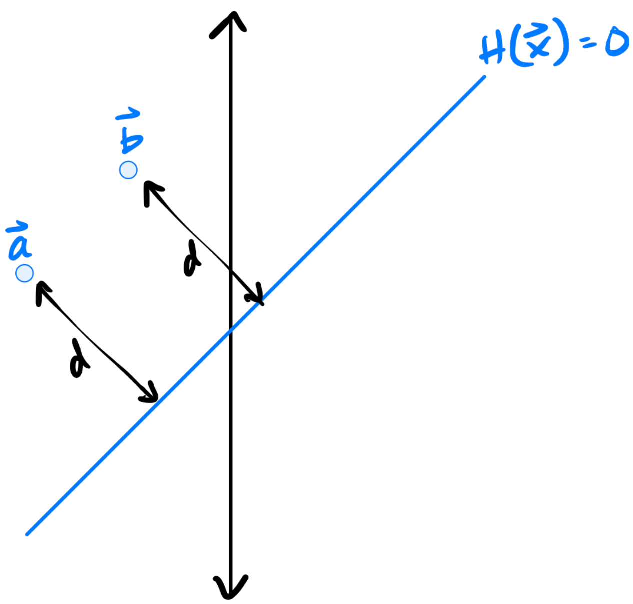
Suppose H_1 and H_2 are two linear prediction rules. Suppose that, on a binary classification data set $\{(\vec{x}^{(1)}, y_1), \ldots, (\vec{x}^{(n)}, y_n)\}$, H_2 achieves a higher training accuracy than H_1 (that is, $\text{sign}(H_2(\vec{x}^{(i)})) = y_i$ for more points than H_1).

True or False: the mean squared error of H_2 must be less than the mean squared error of H_1.

- ◯ True
- ◯ False

Problem 6.

Let $H(\vec{x}) = \mathbb{R}^2 \to \mathbb{R}$ be a linear prediction rule, and suppose two points, $\vec{a}$ and $\vec{b}$, are equidistant from H's decision boundary (the line where $H(\vec{x}) = 0$), as shown in the image below:



True or False: $H(\vec{a}) = H(\vec{b})$.

- ◯ True
- ◯ False

Problem 7.

Does standardizing features affect the output of a linear model?

More precisely, let $\mathcal{X} = \{(\vec{x}^{(1)}, y_1), (\vec{x}^{(2)}, y_2), \ldots, (\vec{x}^{(n)}, y_n)\}$ be a labeled dataset, and let $\mathcal{Z} = \{(\vec{z}^{(1)}, y_1), (\vec{z}^{(2)}, y_2), \ldots, (\vec{z}^{(n)}, y_n)\}$ be the dataset obtained by standardizing each feature in $\mathcal{X}$ (that is, each feature is scaled to have unit standard deviation and zero mean). Note that the y_i's are not standardized – they are the same in both datasets.

Suppose $H_{\mathcal{X}}(\vec{x})$ is a linear predictor trained by minimizing mean squared error on $\mathcal{X}$, and $H_{\mathcal{Z}}(\vec{z})$ is a linear predictor trained by minimizing mean squared error on $\mathcal{Z}$.

True or False: it must be the case that $H_{\mathcal{X}}(\vec{x}^{(i)}) = H_{\mathcal{Z}}(\vec{z}^{(i)})$ for all $i \in \{1, \ldots, n\}$.

- ◯ True
- ◯ False

Problem 8. (2 points)

For this problem, consider the "triangle" basis function, defined by

$$\phi(x; c) = \begin{cases} 1 - |x - c|, & |x - c| < 1, \\ 0 & \text{otherwise.} \end{cases}$$

The c is a parameter of this basis function, determining its "center".

Suppose 3 triangle basis functions, ϕ_1, ϕ_2, ϕ_3, with centers 1, 4, and 5, respectively, are used to map the data to feature space, where a linear classifier H_f is trained. In feature space, the linear predictor has the equation

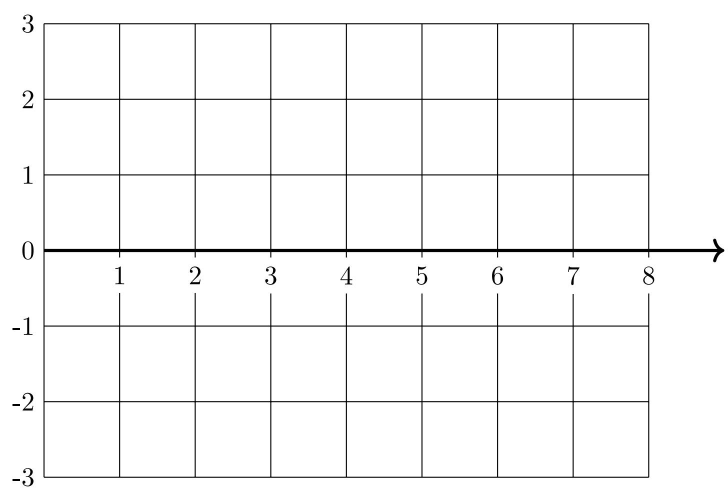
$$H_f(\vec{z}) = 2z_1 - z_2 + 3z_3,$$

where $\vec{z} = (z_1, z_2, z_3)^T$. z_1 corresponds to ϕ_1, z_2 to ϕ_2, and so on.

a) What is the representation of $x = 4.5$ in the feature space?

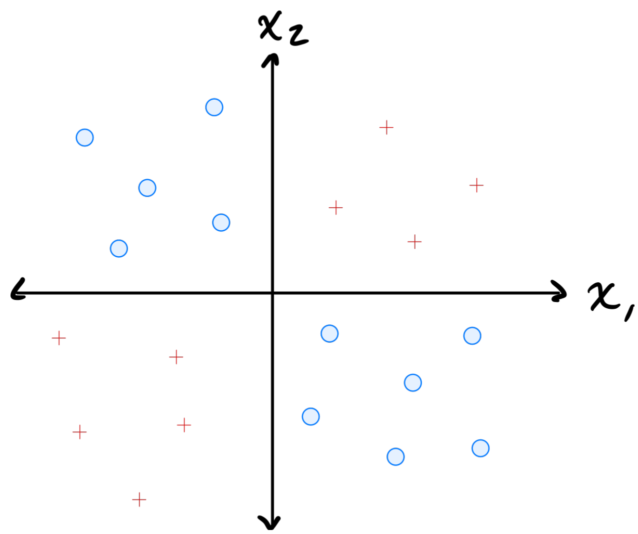
b) Let $H(x)$ be the prediction function in the *original* space. What is $H(4.5)$?

c) Plot $H(x)$ (the prediction function in the original space) from 0 to 8 on the grid below:



Problem 9.

Consider the data shown below:



The data comes from two classes: $\circ$ and $+$.

Suppose a single basis function will be used to map the data to feature space where a linear classifier will be trained. Which of the below is the best choice of basis function?

- ◯ $\phi(x_1, x_2) = \max\{x_1, x_2\}$
- ◯ $\phi(x_1, x_2) = x_1 + x_2$
- ◯ $\phi(x_1, x_2) = x_1^2 + x_2^2$
- ◯ $\phi(x_1, x_2) = x_1 \cdot x_2$

Problem 10.

Let $\mathcal{X} = \{(\vec{x}^{(1)}, y_1), (\vec{x}^{(2)}, y_2), \ldots, (\vec{x}^{(100)}, y_{100})\}$ be a dataset of 100 points, where each feature vector $\vec{x}^{(i)} \in \mathbb{R}^{50}$. Suppose a Gaussian RBF network is trained using 25 Gaussian basis functions.

Recall that a Gaussian RBF network can be viewed as mapping the data to feature space, where a linear prediction rule is trained. In the above scenario, what is the dimensionality of this feature space?

Problem 11.

Suppose a Gaussian RBF network $H(\vec{x}) : \mathbb{R}^2 \to \mathbb{R}$ is trained on a binary classification dataset using four Gaussian RBFs located at:

- $\vec{\mu}_1 = (0, 0)$
- $\vec{\mu}_2 = (1, 0)$
- $\vec{\mu}_3 = (0, 1)$
- $\vec{\mu}_4 = (1, 1)$

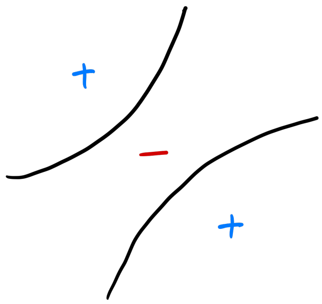
All four RBFs used the same width parameter, σ.

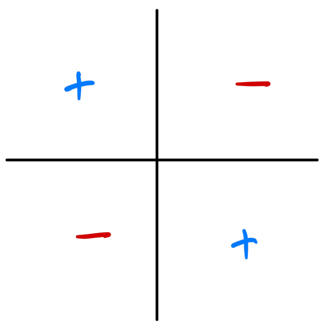
After training, it is found that the weights corresponding to the four basis functions are:

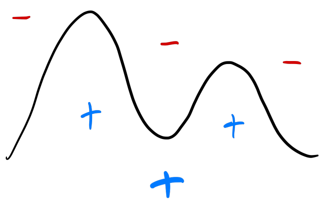
- $w_1 = -3$
- $w_2 = 3$

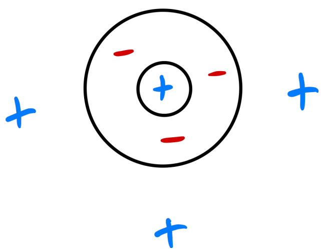
- $w_3 = 3$
- $w_4 = -3$

Which of the below could show the decision boundary of H? The plusses and minuses show where the prediction function, H, is positive and negative.

○ 

○ 

○ 

○ 

Problem 12.

Does standardizing features affect the output of a Gaussian RBF network?

More precisely, let $\mathcal{X} = \{(\vec{x}^{(1)}, y_1), (\vec{x}^{(2)}, y_2), \ldots, (\vec{x}^{(n)}, y_n)\}$ be a dataset, and let $\mathcal{Z} = \{(\vec{z}^{(1)}, y_1), (\vec{z}^{(2)}, y_2), \ldots, (\vec{z}^{(n)}, y_n)\}$ be the dataset obtained by standardizing each feature in $\mathcal{X}$ (that is, each feature is scaled to have unit standard deviation and zero mean). Note that the y_i's are not standardized – they are the same in both datasets.

Suppose $H_{\mathcal{X}}(\vec{x})$ is a Gaussian RBF network trained by minimizing mean squared error on $\mathcal{X}$, and $H_{\mathcal{Z}}(\vec{z})$ is a Gaussian RBF network trained by minimizing mean squared error on $\mathcal{Z}$. Both RBF networks use the same width parameters, σ. The centers used in $H_{\mathcal{Z}}$ are obtained by standardizing the centers used in $H_{\mathcal{X}}$ using the mean and standard deviation of the data, $\vec{x}^{(i)}$.

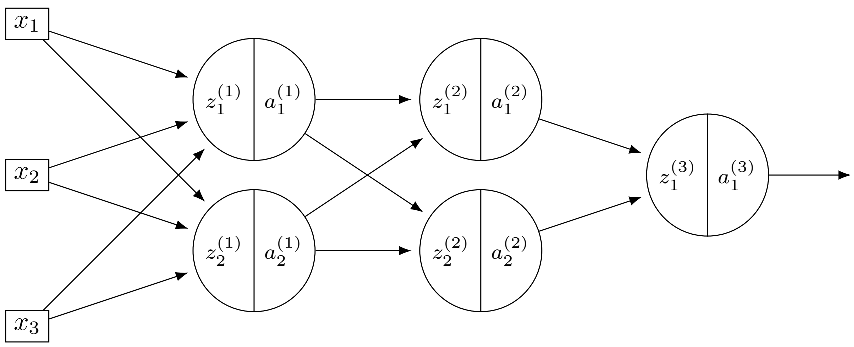
True or False: it must be the case that $H_{\mathcal{X}}(\vec{x}^{(i)}) = H_{\mathcal{Z}}(\vec{z}^{(i)})$ for all $i \in \{1, \ldots, n\}$.

○ True

○ False

Problem 13. (2 points)

Consider the neural network $H(\vec{x})$ shown below:



Let the weights of the network be:

$$W^{(1)} = \begin{pmatrix} 3 & 2 \\ -1 & 2 \\ -1 & 4 \end{pmatrix} \quad W^{(2)} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} \quad W^{(3)} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

Assume that all nodes use **linear** activation functions, and that all biases are zero.

Suppose $\vec{x} = (2, 0, -2)^T$.

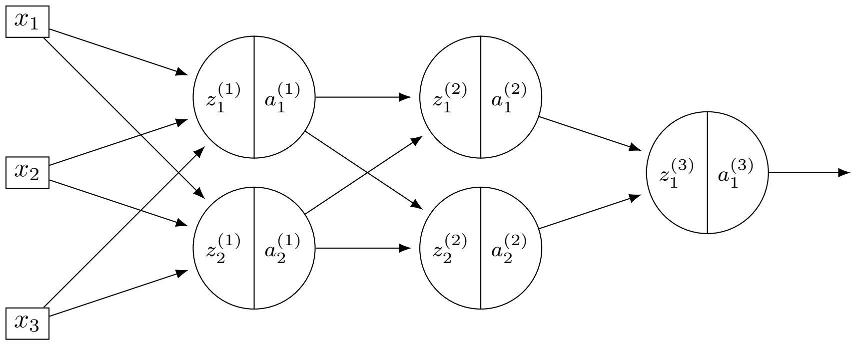
a) What is $a_1^{(1)}$?

b) What is $a_2^{(2)}$?

c) What is $H(\vec{x})$?

Problem 14. (2 points)

Consider the neural network $H(\vec{x})$ shown below:



Let the weights of the network be (as before):

$$W^{(1)} = \begin{pmatrix} 3 & 2 \\ -1 & 2 \\ -1 & 4 \end{pmatrix} \quad W^{(2)} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} \quad W^{(3)} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

Assume that all hidden nodes use **ReLU activation** functions, that the output node uses a linear activation, and that all biases are zero.

Suppose $\vec{x} = (2, 0, -2)^T$.

a) What is $a_1^{(1)}$?

b) What is $a_2^{(2)}$?

c) What is $H(\vec{x})$?

Problem 15.

In the following, when we say a function $f(x)$ "looks like a ReLU", we mean that it has a flat part where $f(x) = 0$, and a linear part with a slope (not necessarily equal to one).

Suppose a deep neural network $H(x) : \mathbb{R} \to \mathbb{R}$ uses ReLU activations in all of its hidden layers.

a) Suppose in addition that the output node uses a linear activation. True or False: when plotted, $H(x)$ must look like a ReLU.

- ○ True
- ○ False

b) Suppose instead that the output node *also* uses a sigmoid activation. True or False: when plotted, $H(x)$ must look like a ReLU.

- ○ True
- ○ False

Problem 16.

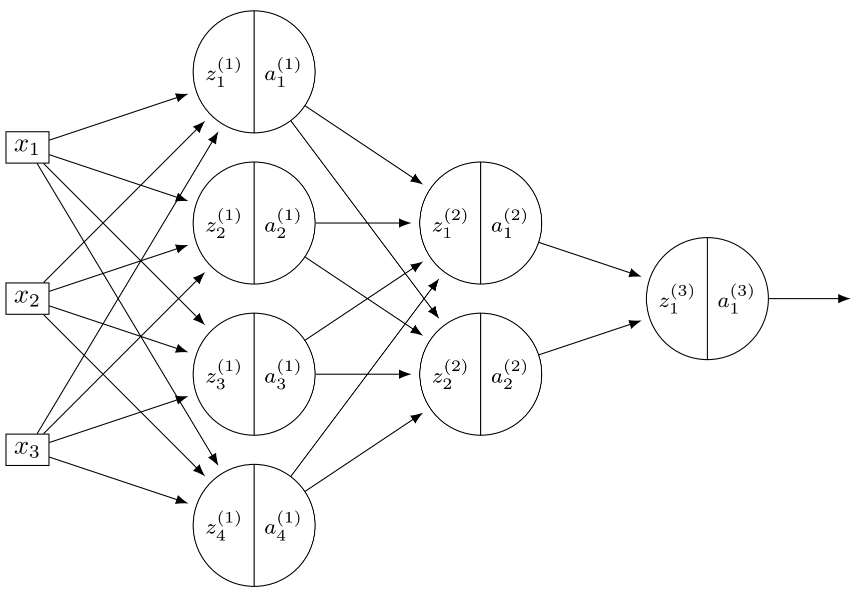
Suppose a deep network $H(\vec{x})$ uses a sigmoid activation in its output layer. Suppose as well that for a given input, $\vec{x}$, $H(\vec{x}) \approx 1$.

True or False: evaluated at this point, $\vec{x}$, $\partial H / \partial W_{ij}^{(\ell)} \approx 0$ for all i, j, ℓ.

- ◯ True
- ◯ False

Problem 17.

Suppose H is the neural network shown below:

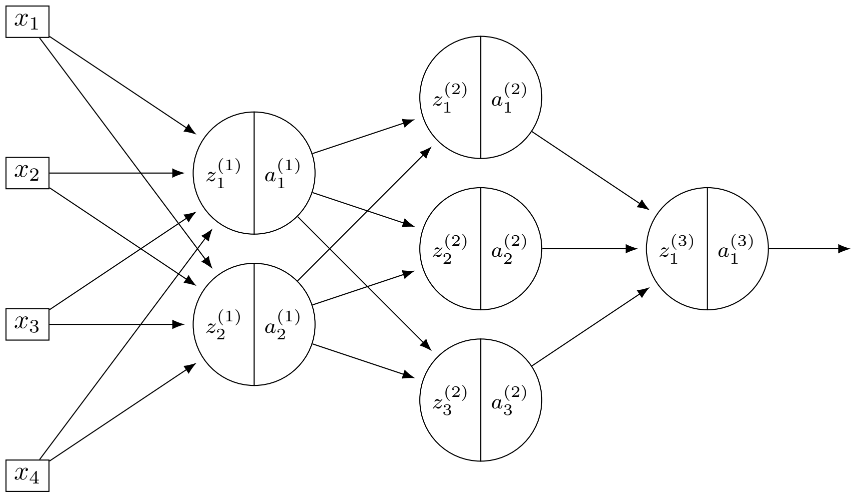


You may assume that all hidden and output nodes have a bias, but that the bias is just not drawn for simplicity of the picture.

The gradient of H with respect to the parameters is a vector. What will be this vector's dimensionality?

Problem 18.

Consider the neural network $H(\vec{x})$ shown below:

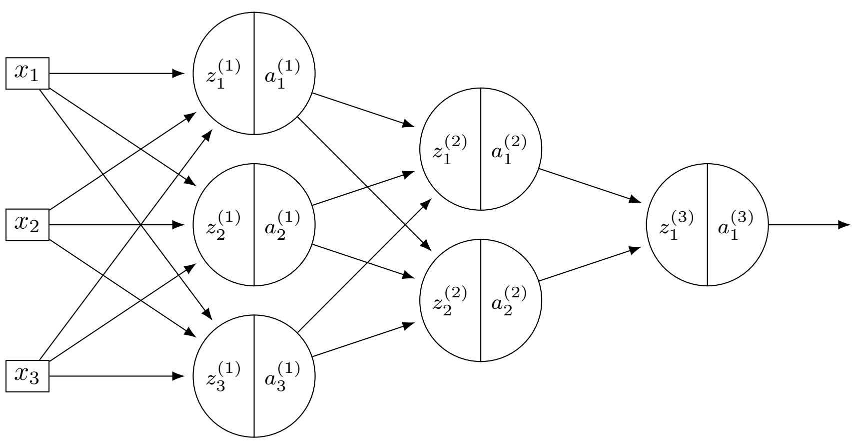


The first layer of this neural network can be thought of as a function $f : \mathbb{R}^d \to \mathbb{R}^k$ mapping feature vectors to a new representation. What are d and k in this case?

d: $\boxed{}$ k: $\boxed{}$

Problem 19. (2 points)

Consider the network shown $H(\vec{x})$ shown below.



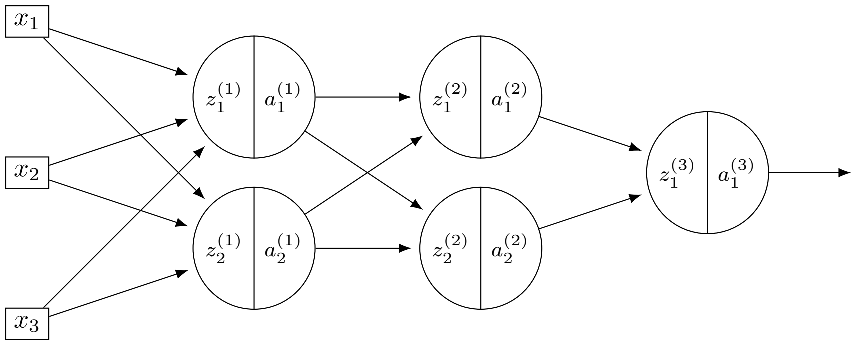
Suppose that the ReLU is used on all hidden nodes, and the linear activation is used on the output node. There are no biases.

Fill in the blank without making reference to any derivatives (apart from g', which is the derivative of the ReLU.

$$\frac{\partial H}{\partial W_{12}^{(1)}} = x_1 g'(z_2^{(1)}) \left(W_{21}^{(2)}\, g'(z_1^{(2)})\, W_{11}^{(3)} + \boxed{} \right)$$

Problem 20. (3 points)

Consider the neural network $H(\vec{x})$ shown below:



Assume that all hidden nodes use **ReLU activation** functions, that the output node uses a linear activation, and that there are no biases.

Let the weights of the second layer of the network be: $W^{(2)} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}$.

In addition, suppose it is known that, when the network evaluated at $\vec{x}$ with weight vector $\vec{w}$:

$$z_1^{(1)} = 3, \qquad z_2^{(1)} = -2, \qquad \partial H/\partial z_1^{(2)} = -3, \qquad \partial H/\partial z_2^{(2)} = 1$$

a) What is $\partial H/\partial a_1^{(1)}$, as a number?

b) What is $\partial H/\partial a_2^{(1)}$, as a number?

c) What is $\partial H/\partial z_1^{(1)}$, as a number?

d) What is $\partial H/\partial z_2^{(1)}$, as a number?

e) What is $\partial H/\partial W_{11}^{(2)}$, as a number?

f) What is $\partial H/\partial W_{21}^{(2)}$, as a number?

Before turning in your exam, please check that your name is on every page.

(You may use this area for scratch work.)